Questions

Q1.

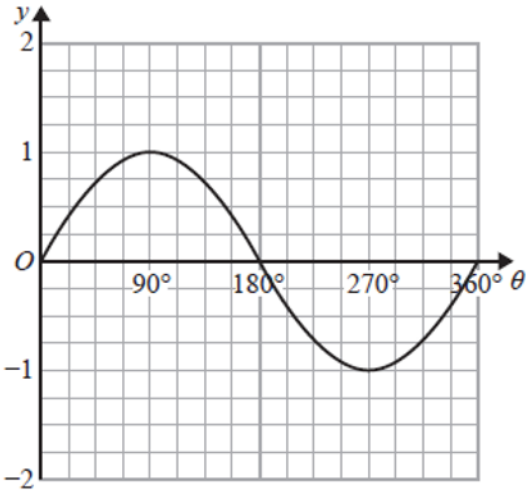


Figure 3

Figure 3 shows a plot of the curve with equation $y = \sin\theta$, $0 \leq \theta \leq 360°$

(a) State the coordinates of the minimum point on the curve with equation

$y = 4\sin\theta$, $0 \leq \theta \leq 360°$

(2)

A copy of Figure 3, called Diagram 1, is shown here.

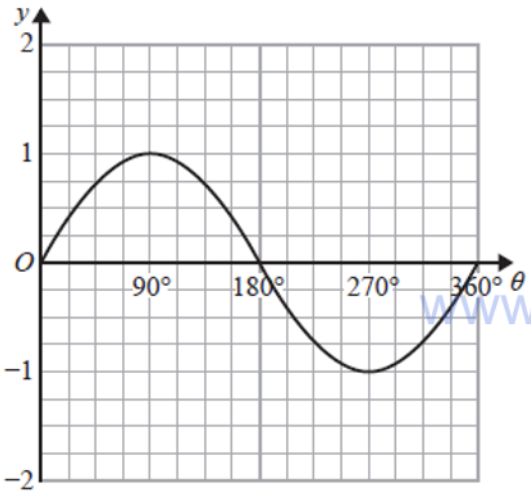


Diagram 1

(b) On Diagram 1, sketch and label the curves

(i) $y = 1 + \sin\theta$, $0 \leq \theta \leq 360°$

(ii) $y = \tan\theta$, $0 \leq \theta \leq 360°$

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(2)

(c) Hence find the number of solutions of the equation

(i) $\tan\theta = 1 + \sin\theta$ that lie in the region $0 \leq \theta \leq 2160°$

(i) $\tan\theta = 1 + \sin\theta$ that lie in the region $0 \leq \theta \leq 1980°$

(3)

(Total for question = 7 marks)

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(Q09 WMA11/01, June 2019)

Q2.

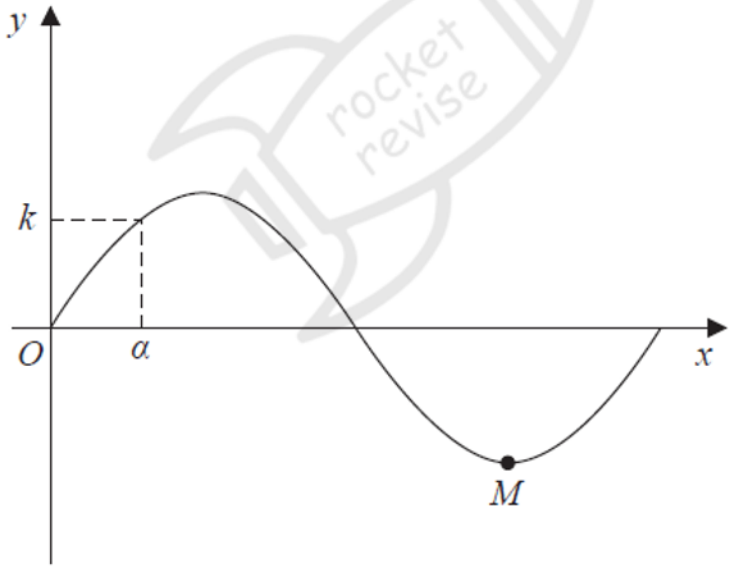


Figure 5

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Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C.

(a) State the period of C.

(1)

(b) State the coordinates of M.

(1)

The smallest positive solution of the equation

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$\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α.

Find, in terms of α,

(c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,

(ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$.

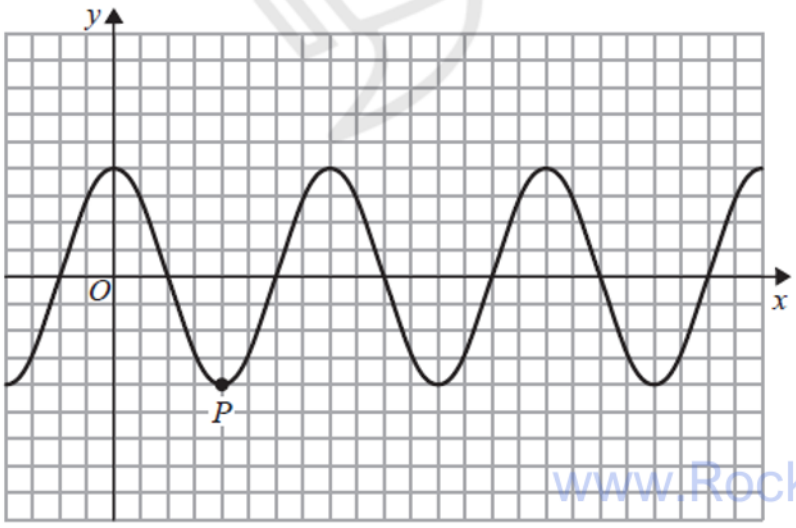
(2)

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(Total for question = 4 marks)

(Q09 WMA11/01, Oct 2019)

Q3.



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Figure 2

Figure 2 shows a plot of part of the curve with equation $y = \cos 2x$ with x being measured in radians.

The point P, shown on Figure 2, is a minimum point on the curve.

(a) State the coordinates of P.

(2)

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A copy of Figure 2, called Diagram 1, is shown at the top of the next page.

(b) Sketch, on Diagram 1, the curve with equation $y = \sin x$

(2)

(c) Hence, or otherwise, deduce the number of solutions of the equation

(i) $\cos 2x = \sin x$ that lie in the region $0 \leq x \leq 20\pi$

(ii) $\cos 2x = \sin x$ that lie in the region $0 \leq x \leq 21\pi$

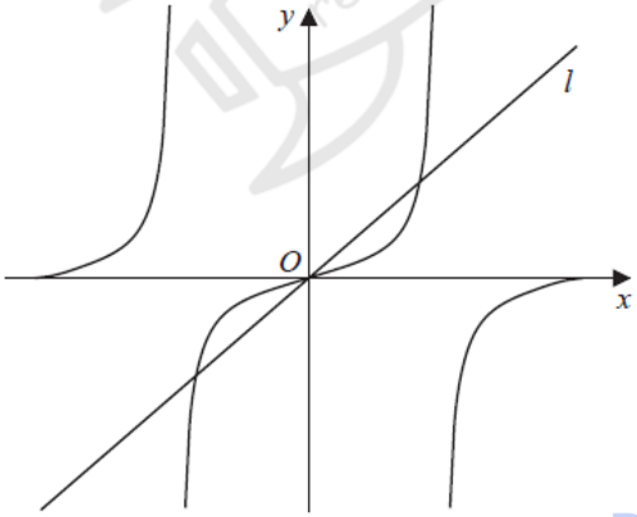
(2)

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(Total for question = 6 marks)

(Q05 WMA11/01, Jan 2019)

Q4.



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Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

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(a) State the period of tan x

(1)

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

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(1)

(Total for question = 4 marks)

(Q09 WMA11/01, Jan 2023)

Q5.

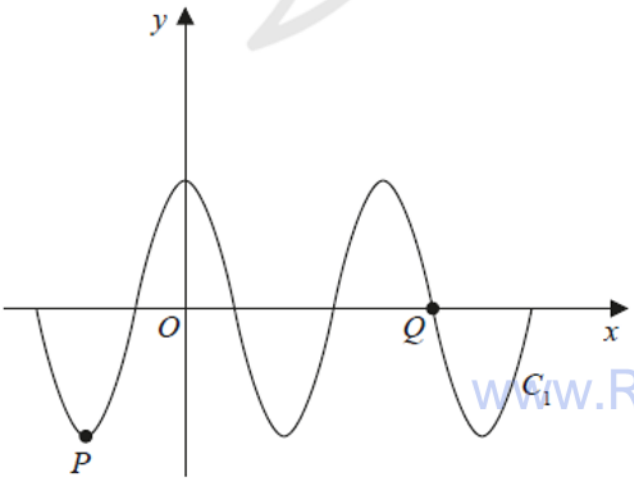


Figure 1

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Figure 1 shows a sketch of part of the curve C_1 with equation $y = 4 \cos x°$

The point P and the point Q lie on C_1 and are shown in Figure 1.

(a) State

(i) the coordinates of P,

(ii) the coordinates of Q.

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(3)

The curve C_2 has equation $y = 4 \cos x° + k$, where k is a constant.

Curve C_2 has a minimum y value of –1

The point R is the maximum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R.

(2)

(Total for question = 5 marks)

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(Q03 WMA11/01, Jan 2021)

Q6.

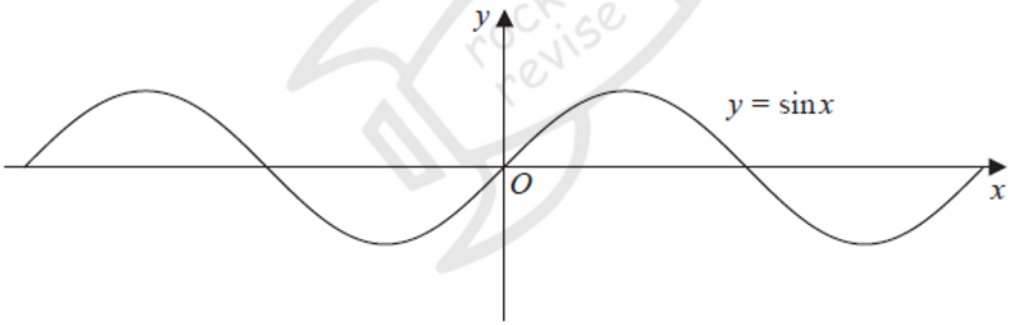


Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin \alpha = p$, where $0 < \alpha < 90°$

(a) state, in terms of p, the value of

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(i) $2 \sin (180° - \alpha)$

(ii) $\sin (\alpha - 180°)$

(iii) $3 + \sin (180° + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown below.

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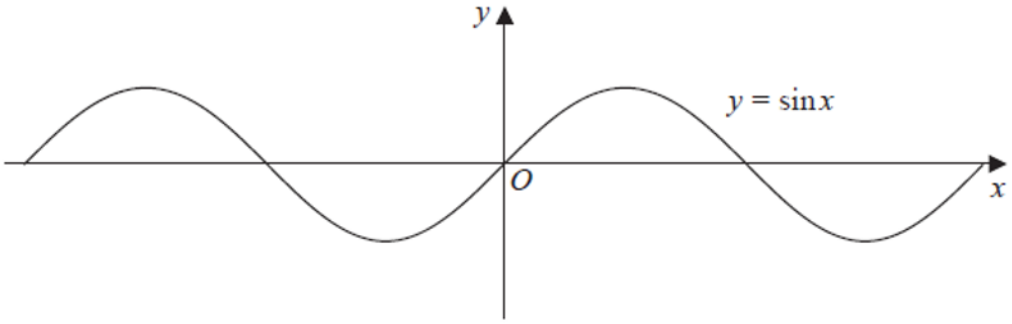


Diagram 1

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

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(2)

(c) Hence find, in terms of α, the x coordinates of any points in the interval $0 < x < 180°$ where

$$\sin 2x = p$$

(3)

(Total for question = 8 marks)

(**Q09 WMA11/01, June 2022**)

Q7.

(i)

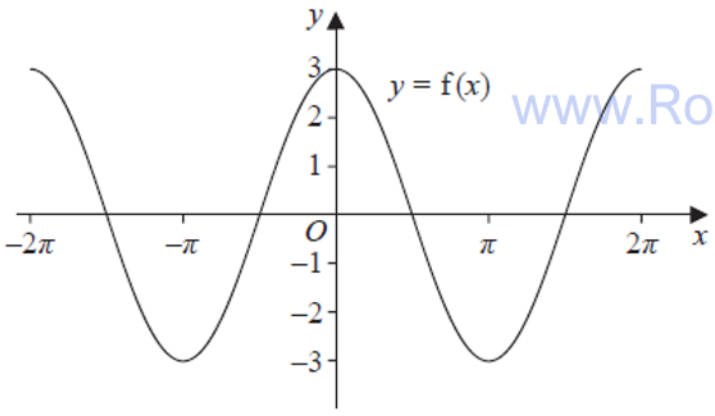


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

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(a) Write down an expression for f(x)

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = \mathrm{f}\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

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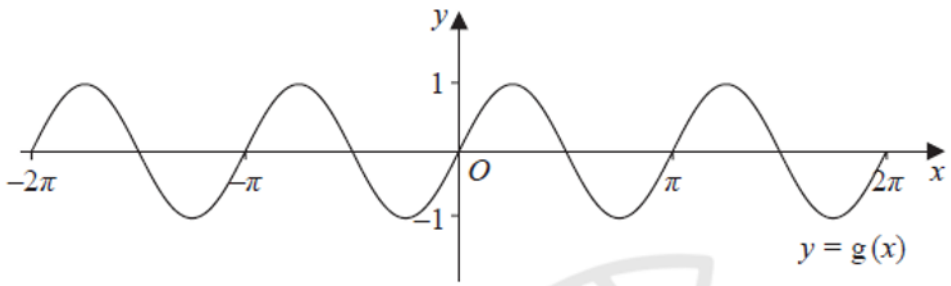


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = \mathrm{g}(x)$

(a) Write down an expression for g(x)

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = \mathrm{g}(x) - 2$

Show clearly the coordinates of the y intercept.

(2)

(Total for question = 9 marks)

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(Q09 WMA11/01, June 2023)

Q8.

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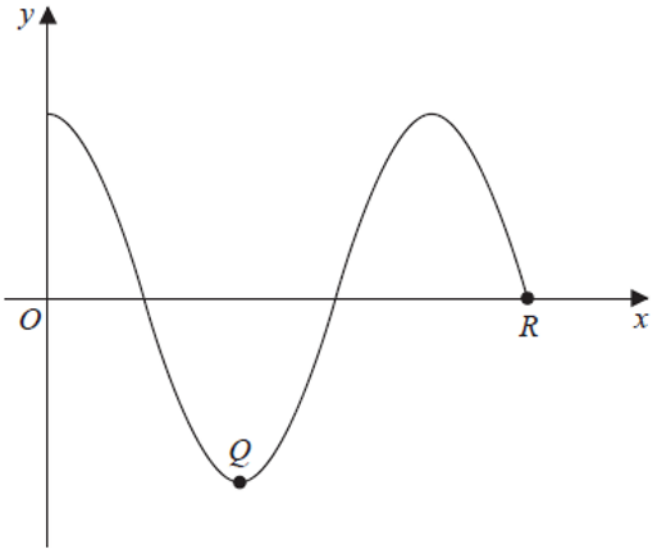


Figure 2

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Figure 2 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \cos 2x° \qquad 0 \leq x \leq k$$

The point Q and the point $R(k, 0)$ lie on the curve and are shown in Figure 2.

(a) State

(i) the coordinates of Q,

(ii) the value of k.

(3)

(b) Given that there are exactly two solutions to the equation

$$\cos 2x° = p \qquad \text{in the region } 0 \leq x \leq k$$

find the range of possible values for p.

(2)

(Total for question = 5 marks)

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(Q04 WMA11/01, Oct 2021)

Q9.

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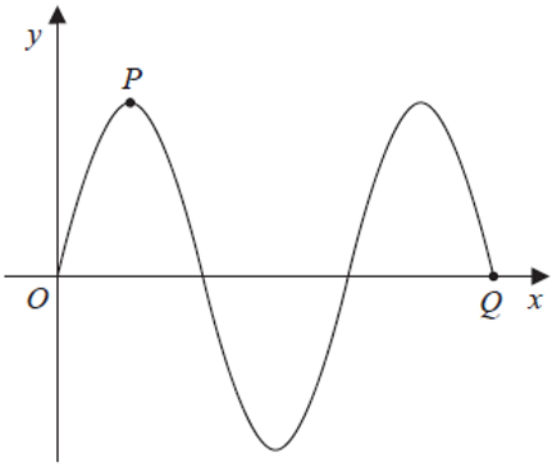


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 3\sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

(i) the coordinates of P,

(ii) the coordinates of Q.

(3)

A different curve C_2 has equation $y = 3\sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R.

(2)

(Total for question = 5 marks)

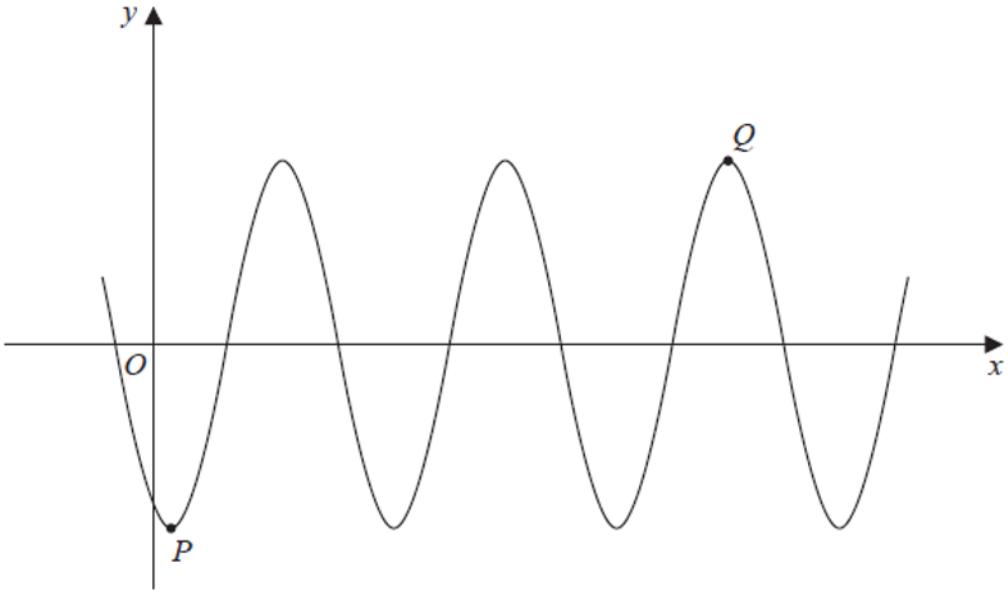
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(Q07 WMA11/01, Jan 2020)

Q10.

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Figure 4

Figure 4 shows part of the curve with equation

$$y = A\cos(x - 30)°$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates (30, –3) as shown in Figure 4.

(a) Write down the value of A.

(1)

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q.

(3)

(Total for question = 4 marks)

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(Q09 WMA11/01, Jan 2022)

Q11.

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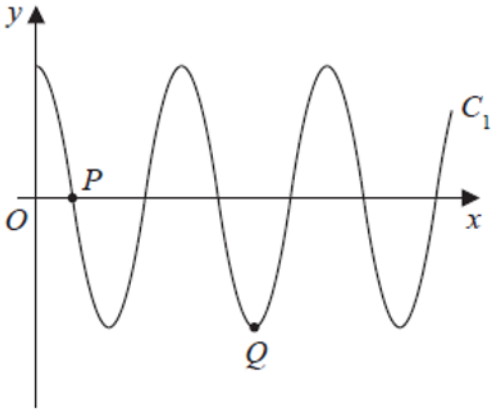


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3\cos\left(\frac{x}{n}\right)^\circ \qquad x \geqslant 0$$

where n is a constant.

The curve C_1 cuts the positive x-axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

The point Q, shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q.

(2)

The curve C_2 has equation $y = 2\sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k.

(2)

(Total for question = 6 marks)

(Q10 WMA11/01, Oct 2023)

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